

ALIE Byzantine Attack Comparison: Different Delta Values (CNN Simplest Sweep)

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1 Introduction

This document presents comparison grids (2x2 subfigures) of the MNIST CNN model under the Optimal A Little Is Enough (ALIE) Byzantine attack. We compare four different step size (δ) parameters:

- **ALIE** $\delta = 10$ (Default positive large step size)
- **ALIE** $\delta = 1$ (Positive small step size)
- **ALIE** $\delta = -1$ (Negative small step size)
- **ALIE** $\delta = -10$ (Negative large step size)

Each comparison illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 5]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$) with a total of $N = 10$ honest clients.

2 ALIE Attack: Best Aggregator Choice

This section compares performance under ALIE using the dynamically selected best overall aggregator.

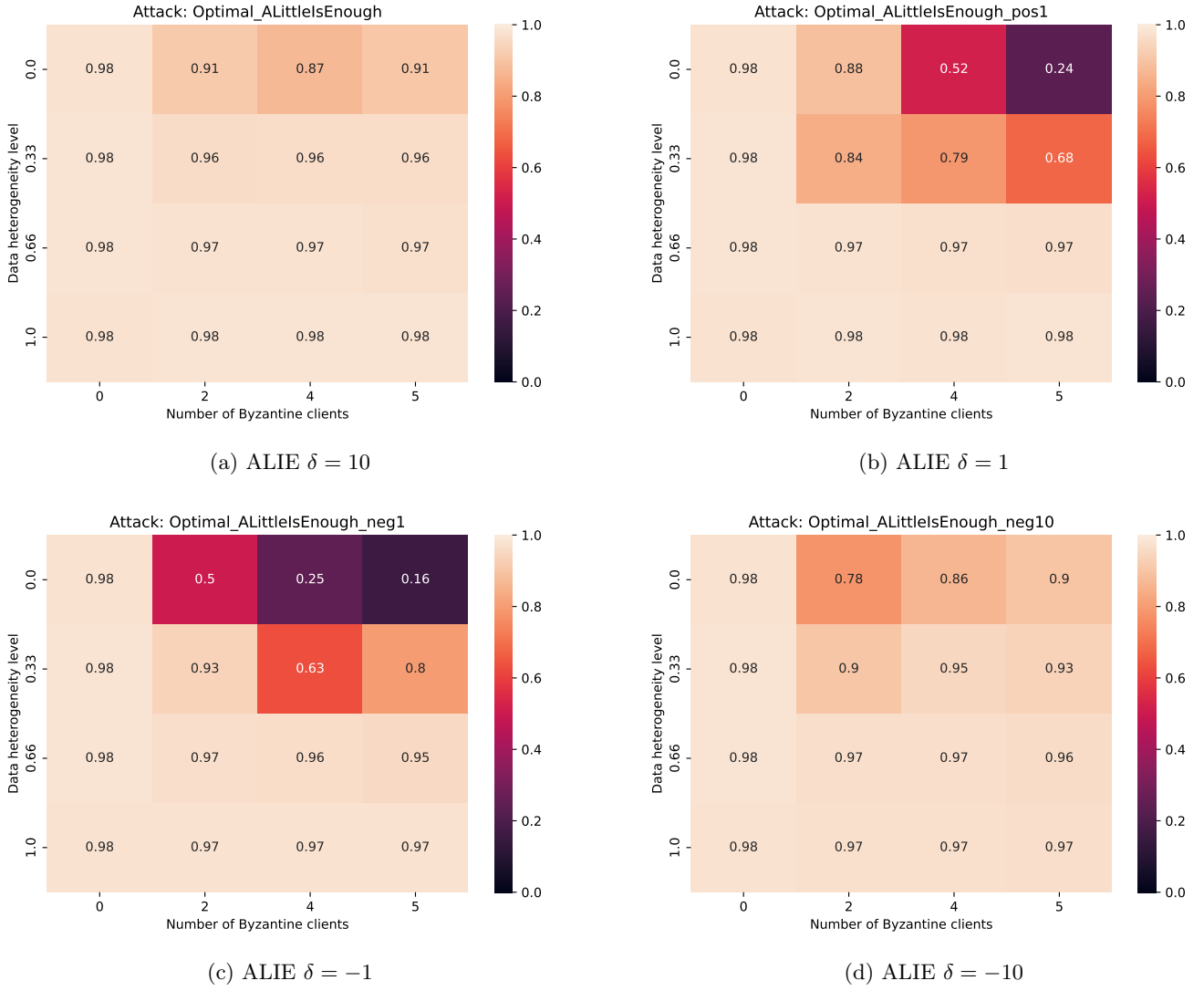
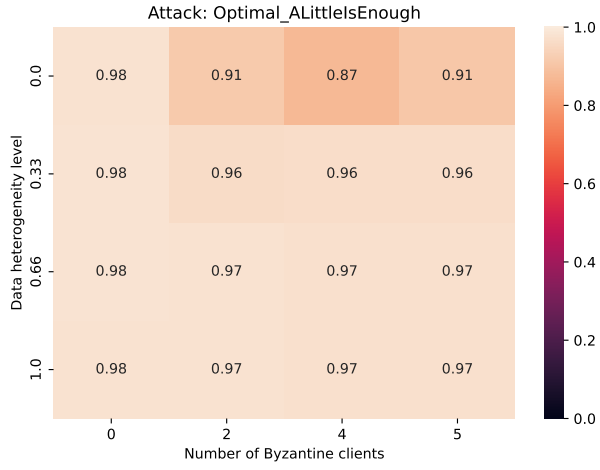


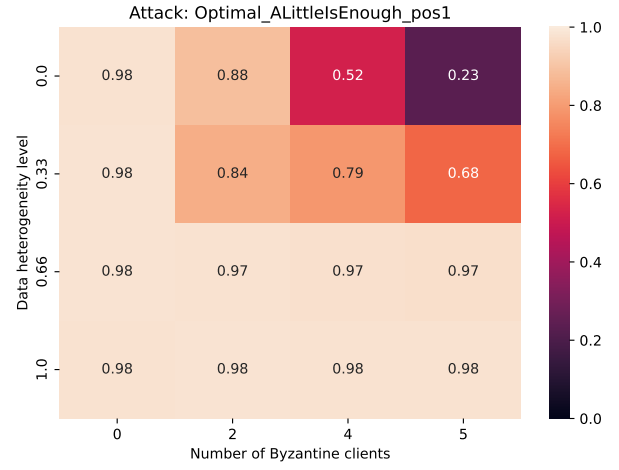
Figure 1: Accuracy comparison under ALIE attacks using the best overall aggregator selection.

3 ALIE Attack: Centered Clipping

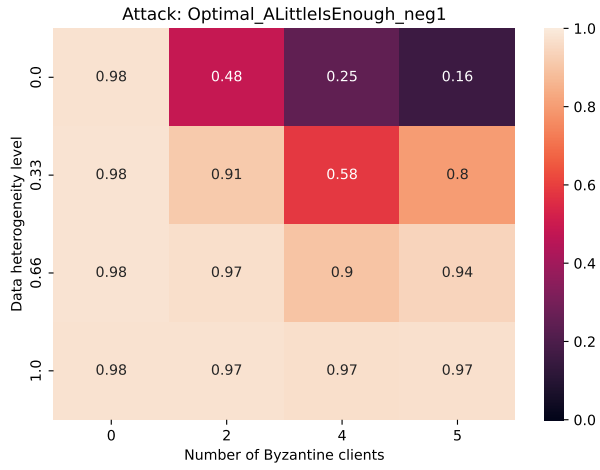
This section compares performance under ALIE using Centered Clipping.



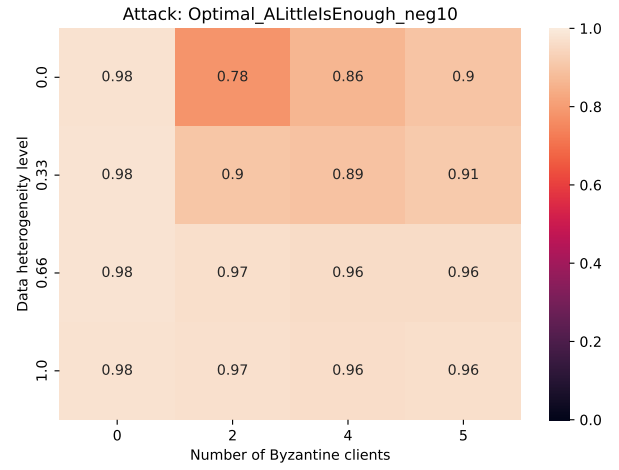
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$

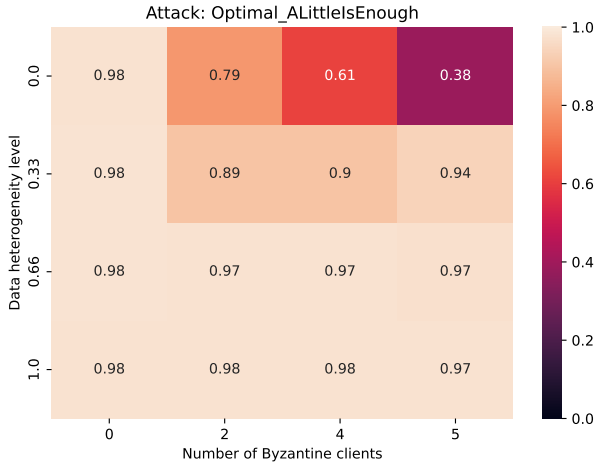


(d) ALIE $\delta = -10$

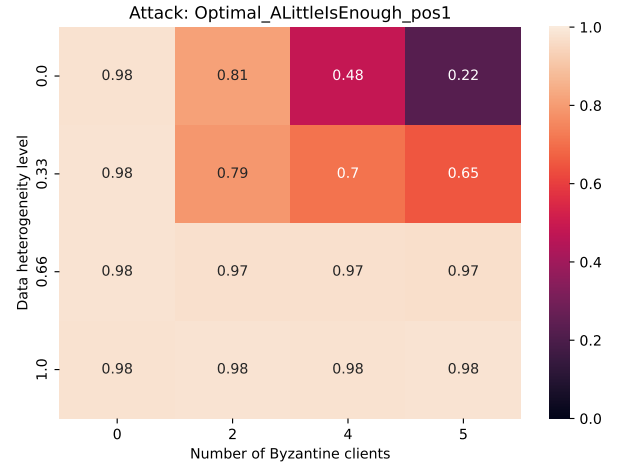
Figure 2: Accuracy comparison under ALIE attacks using Centered Clipping.

4 ALIE Attack: Geometric Median

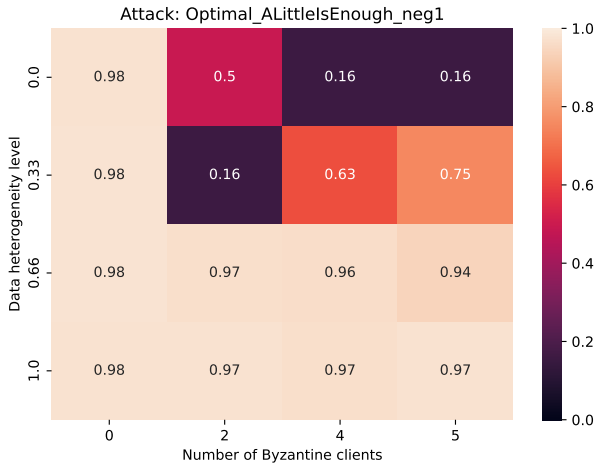
This section compares performance under ALIE using Geometric Median.



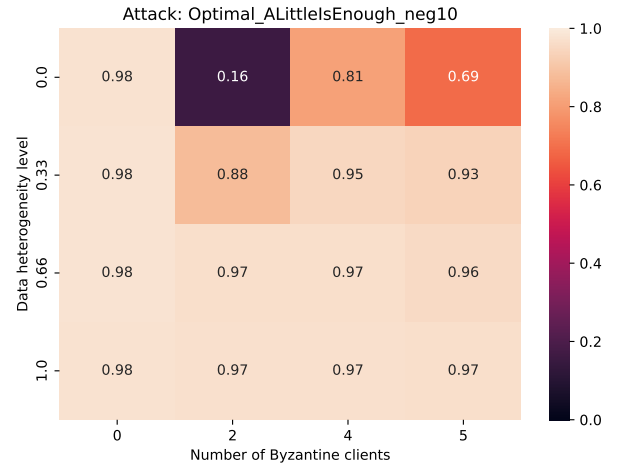
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$

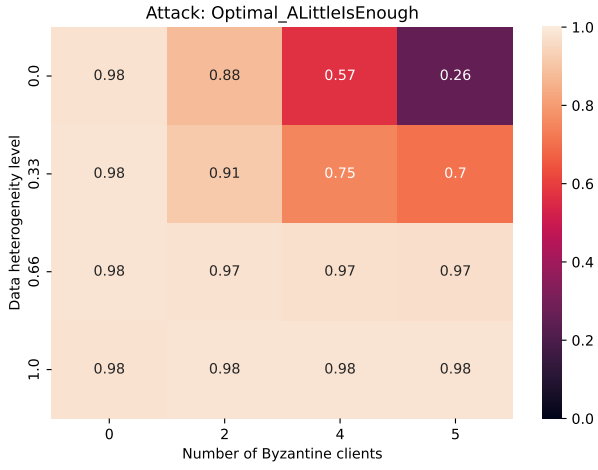


(d) ALIE $\delta = -10$

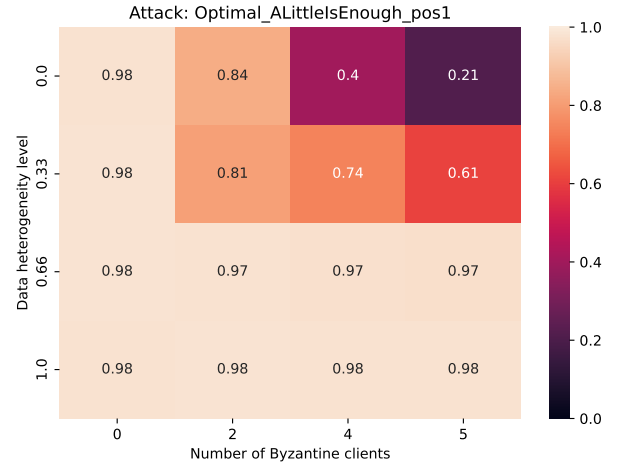
Figure 3: Accuracy comparison under ALIE attacks using Geometric Median.

5 ALIE Attack: Multi-Krum

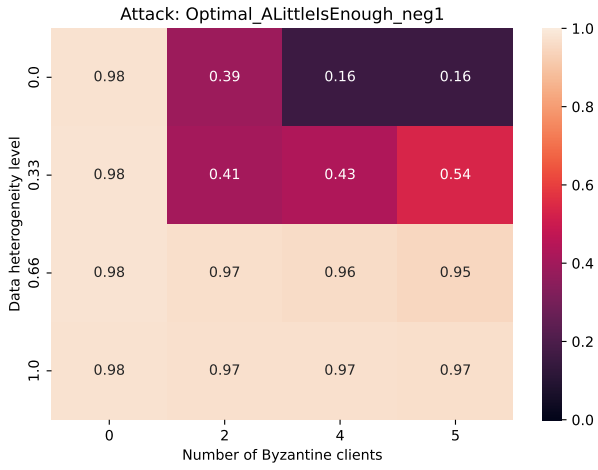
This section compares performance under ALIE using Multi-Krum.



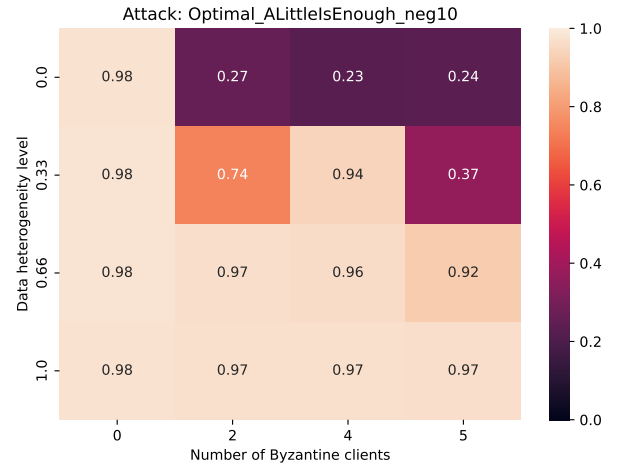
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$

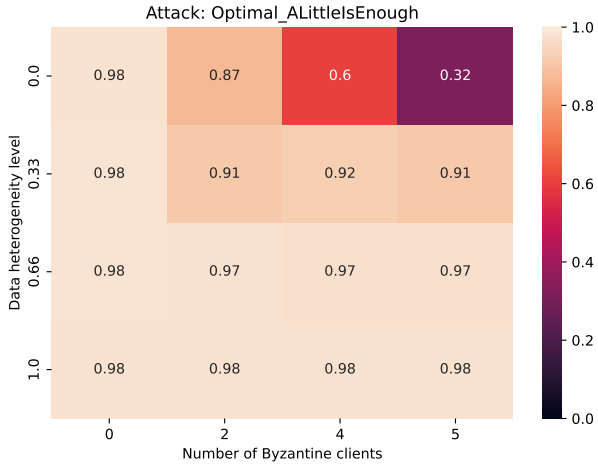


(d) ALIE $\delta = -10$

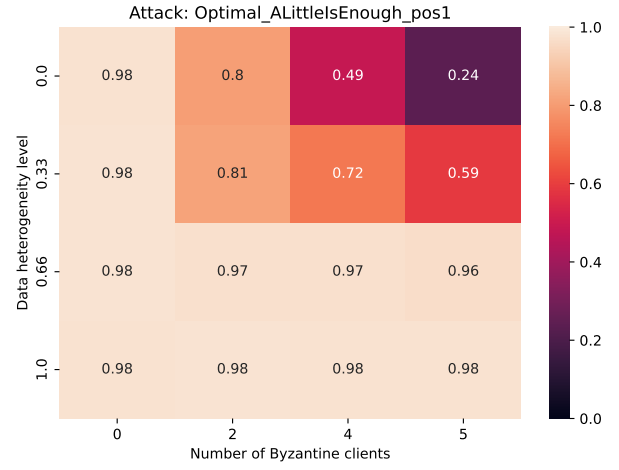
Figure 4: Accuracy comparison under ALIE attacks using Multi-Krum.

6 ALIE Attack: Trimmed Mean

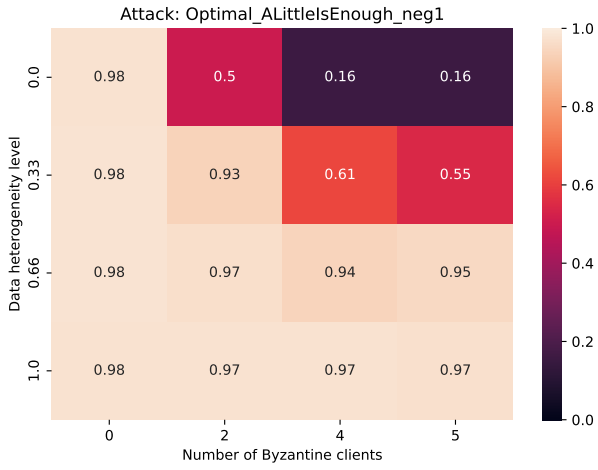
This section compares performance under ALIE using Trimmed Mean.



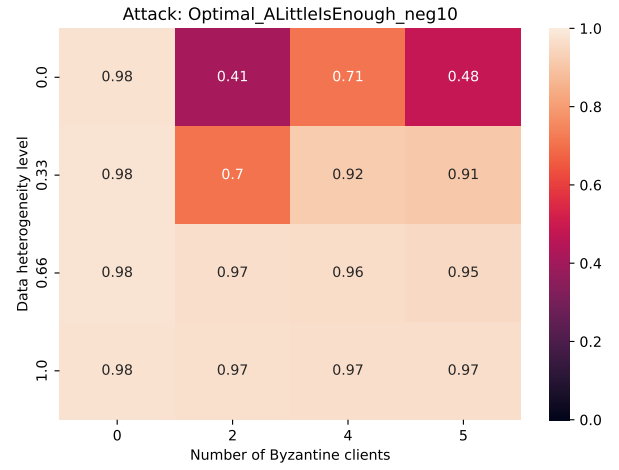
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$



(d) ALIE $\delta = -10$

Figure 5: Accuracy comparison under ALIE attacks using Trimmed Mean.